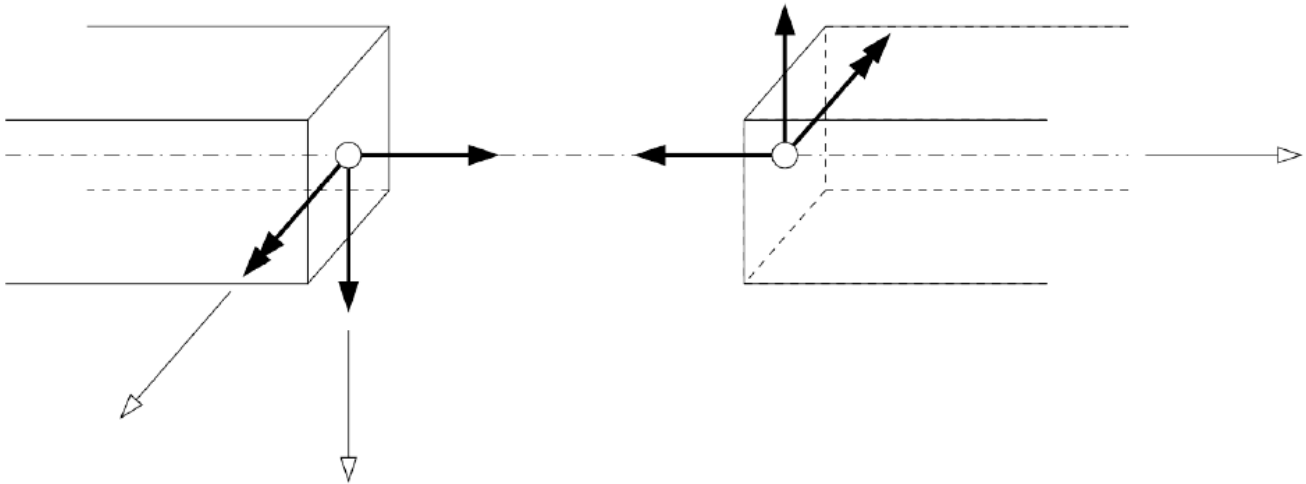
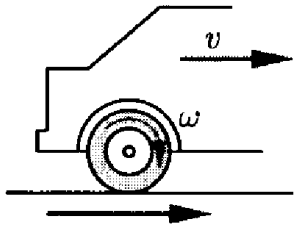
Engineering Mechanics I

Lecture

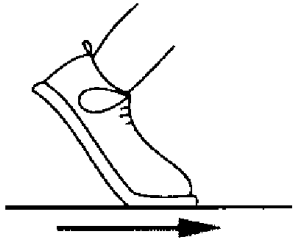


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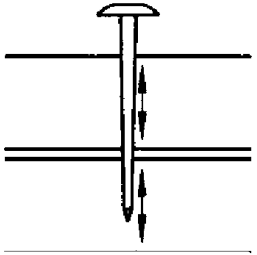
11. Friction



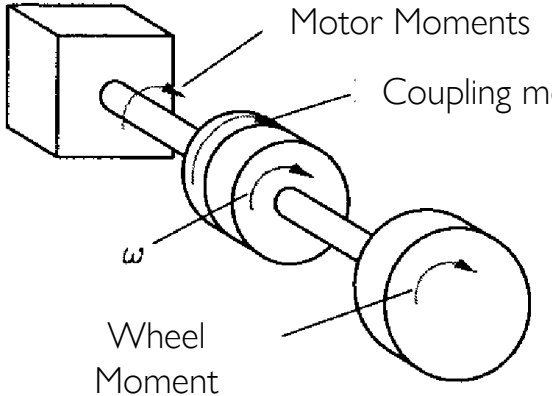
Motive Force

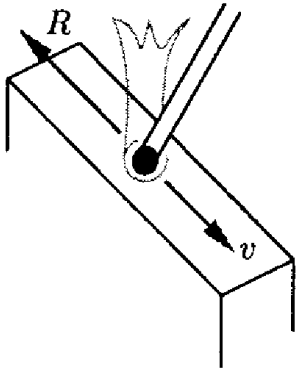


Traction while walking



Force-fit connection





Friction generates heat

Autonomous Aerial Systems

11. Friction

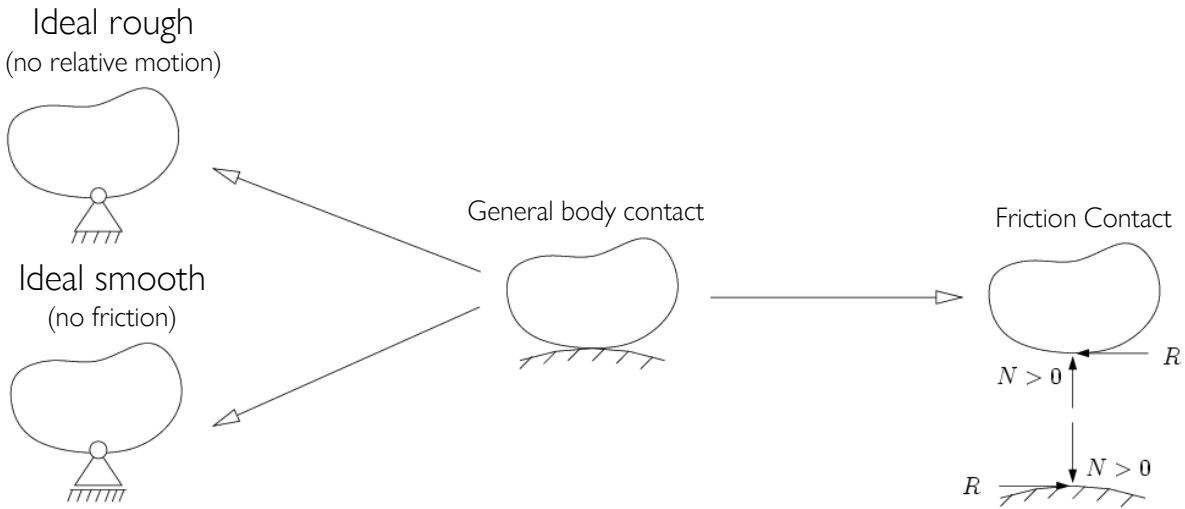
- Ideally smooth contact surfaces: only force transmission perpendicular to the contact surface.
- Real bodies have surface roughness and can therefore also transmit tangential forces.
- Great importance for technical systems.
- Distinction between adhesive contact (static friction) and sliding contact (dynamic friction).

11. Friction

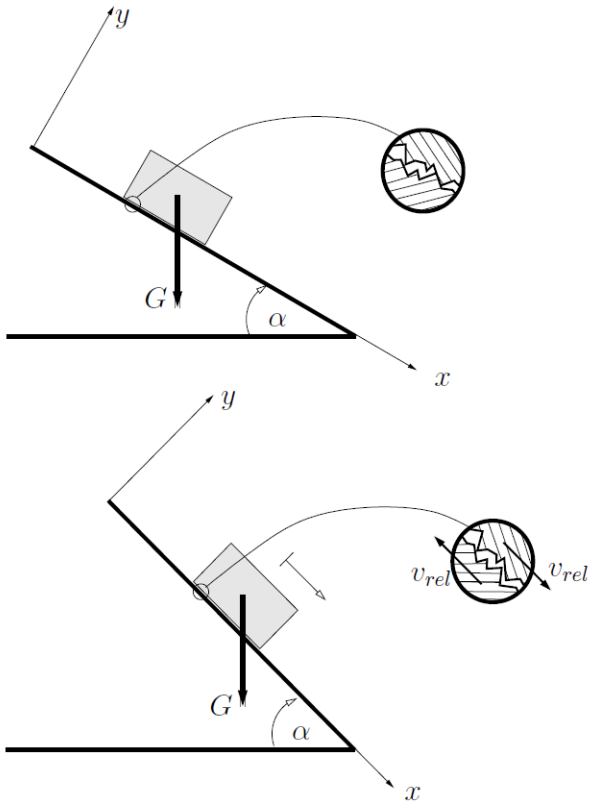
Determination of the **tangential force R**?
Different friction models for complex phenomena:

For example:

- **Dry friction** (Coulomb friction), velocity independent
- **Viscous friction** (damping), velocity dependent (cf. EM 2)



11.1. Mechanics of dry friction



Experiment: Block on inclined plane

- Body at rest, if angle of inclination small angle of friction α_{max}
  $\alpha \leq \alpha_{max}$: Static friction
  (Static Equilibrium)

- Body is gliding if $\alpha > \alpha_{max}$
  $\alpha \geq \alpha_{max}$: Kinetic friction
  (No Static Equilibrium)

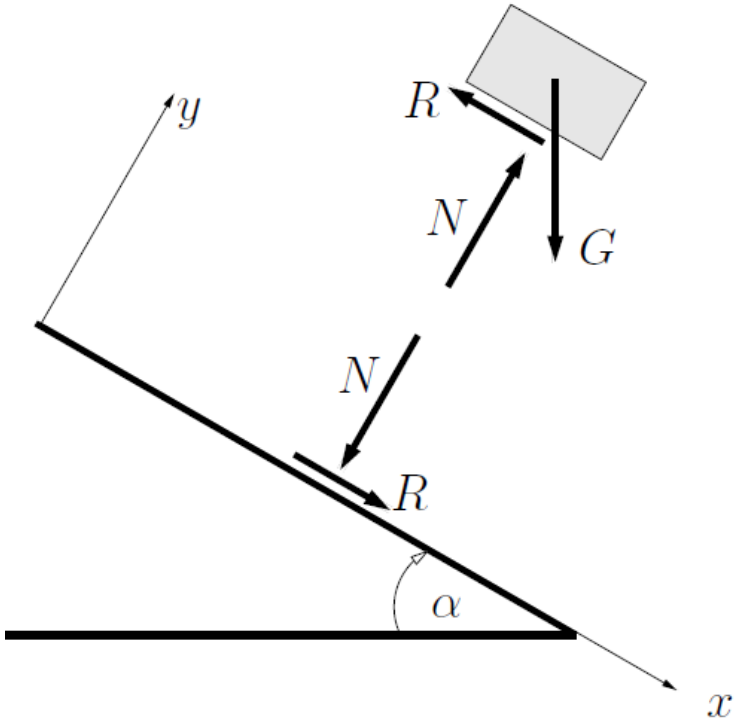
11.2. Static friction

- Block adheres if $\alpha \leq \alpha_{max}$
- Free cutting and balance

$$R = G\sin(\alpha) > 0$$

$$N = G\cos(\alpha) > 0$$

- Direction of action of R always opposite potential direction of motion



11.2. Static friction

- Static or dynamic friction modeled by Coulomb's law.

  Adhesion if:

Box 11.1

- Coefficient of static friction μ_0 depends on materials rubbing against each other.

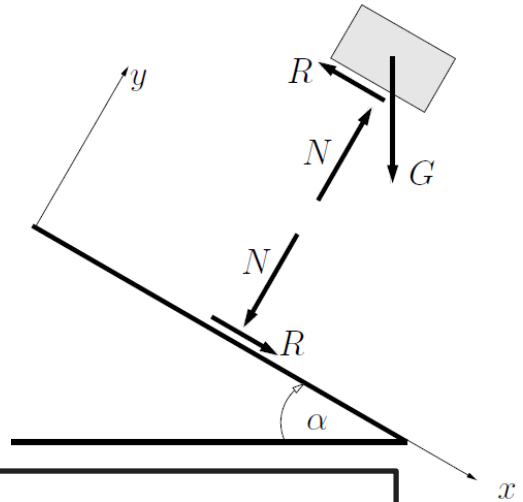
- Adhesion limit: " = "

11.2. Static friction

Block adheres if α_{max}:

- Angle of inclination for which the block just still adheres

$$R = G\sin(\alpha)$$
$$N = G\cos(\alpha)$$
$$|\vec{R}| = \mu_0|\vec{N}| \leftrightarrow R = \mu_0|N|$$

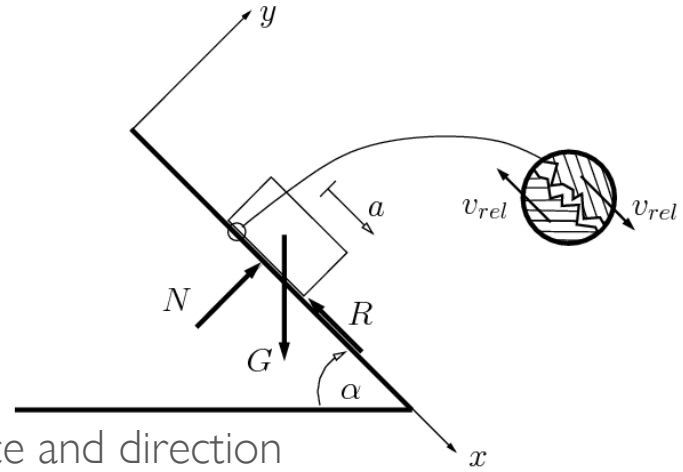


- With given static friction coefficient, friction angle quantifiable:

Box 11.2

- If $\alpha \geq \alpha_{max}$, there is a sudden transition to sliding friction and thus to the movement of the body.

11.3. Kinetic Friction



- In experiment: Block slides if
- No static equilibrium!
- Frictional force dependent on pressure force and direction

$$R = f\left(N, v_{\text{rel}}\right) < G \sin\alpha \quad \text{for } \alpha > \alpha_{\max}$$

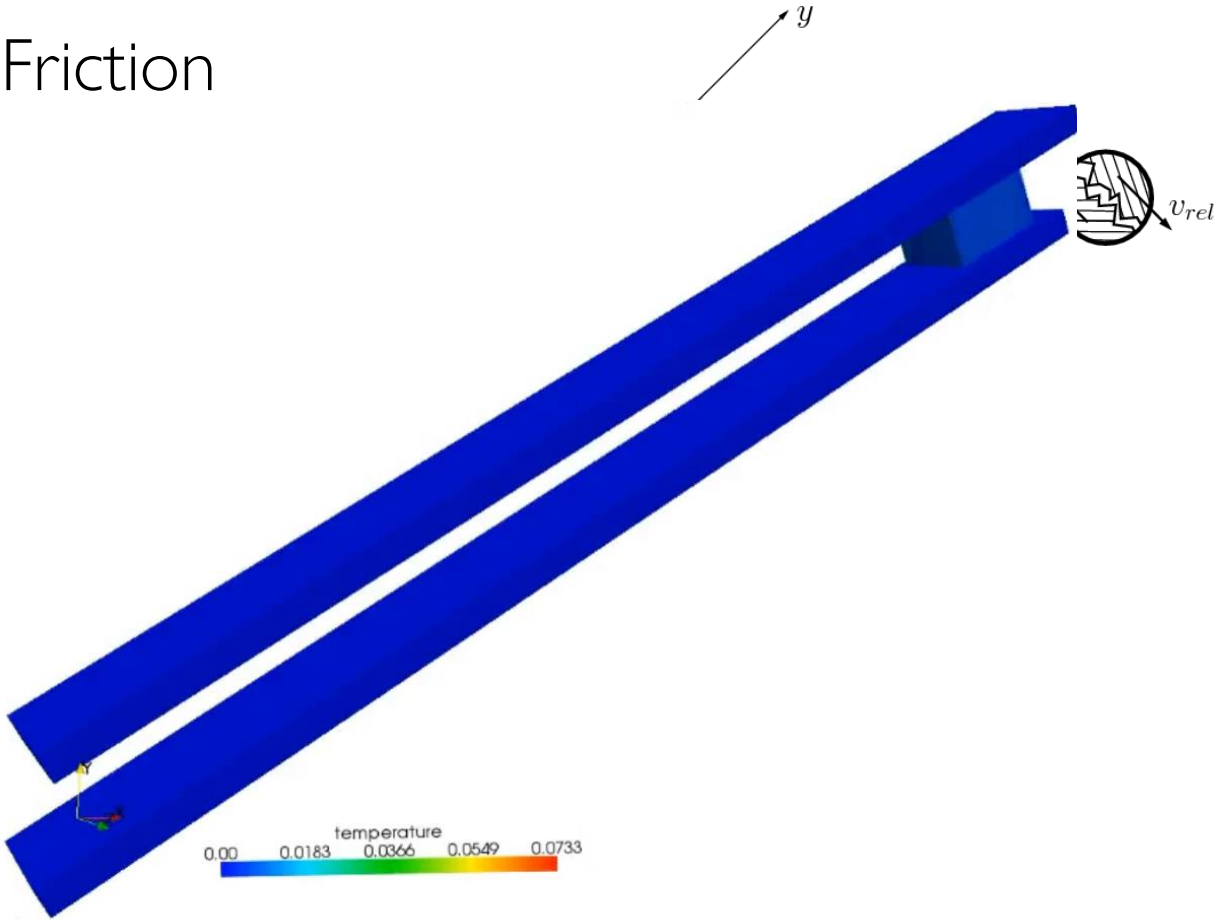
- Work is performed.

11.3. Kinetic Friction

- In experiment
- No static equi
- Frictional force

$$R =$$

- Work is perfo



11.3. Kinetic Friction – Properties of dry kinetic friction

- Sliding friction always acts in the opposite direction to the relative velocity of the sliding surfaces. sliding surfaces.

Box 11.3

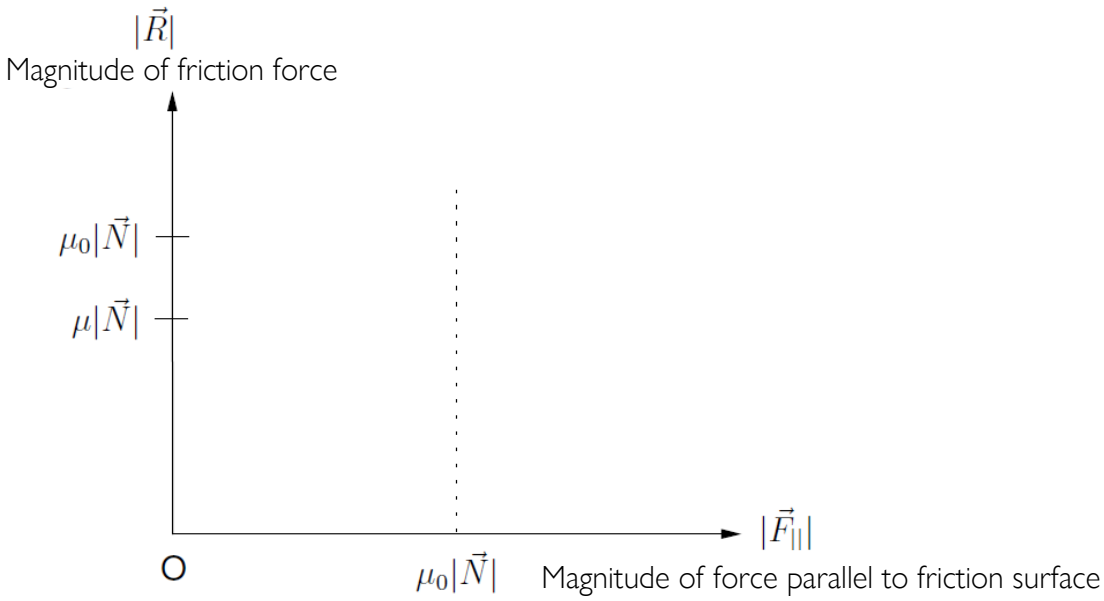
- The amount is independent of the amount of the relative velocity.
- The sliding friction is proportional to the normal force and independent of the size of the contact surface. The law of friction according to Coulomb is:

Box 11.4

With material-dependent coefficient of sliding friction. It holds: $\mu \leq \mu_0$

11.3. Kinetic Friction

- Graphical illustration of the *Coulomb friction curve*:



Box 11.5

11.3. Kinetic Friction

- Summary

Box 11.6

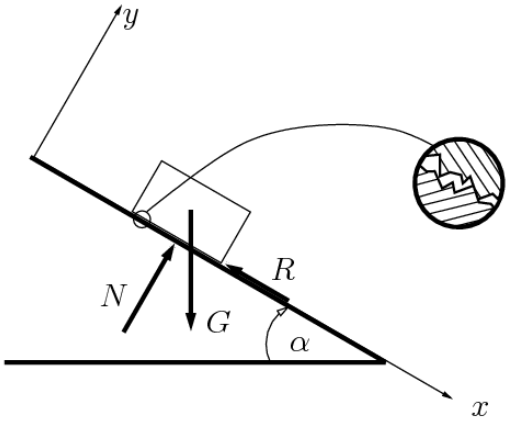
- Friction parameters for common material pairs:

:

Friction pair	Static friction coefficient	Kinetic friction coefficient
Steel - Ice	0.03	0.015
Steel – Steel	0.15 … 0.5	0.1 … 0.4
Wood – Wood	0.5	0.3
Car tire - asphalt	0.7 … 0.9	0.5 … 0.8

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11.4. Cone of adhesion and angle of adhesion



The following relationship results from the previous experiment:

- Up to the limiting case of static friction applies:

$$G \sin(\alpha) \leq \mu_0 G \cos(\alpha)$$

$$\tan(\alpha) \leq \mu_0$$

- The angle between the resulting force on the body and the surface normal surface normal is introduced as ρ (in the example $\rho = \alpha$).

- Thus, angle of adhesion ρ_0 and the corresponding adhesion condition is defined as

$$\tan \rho_0 = \mu_0$$

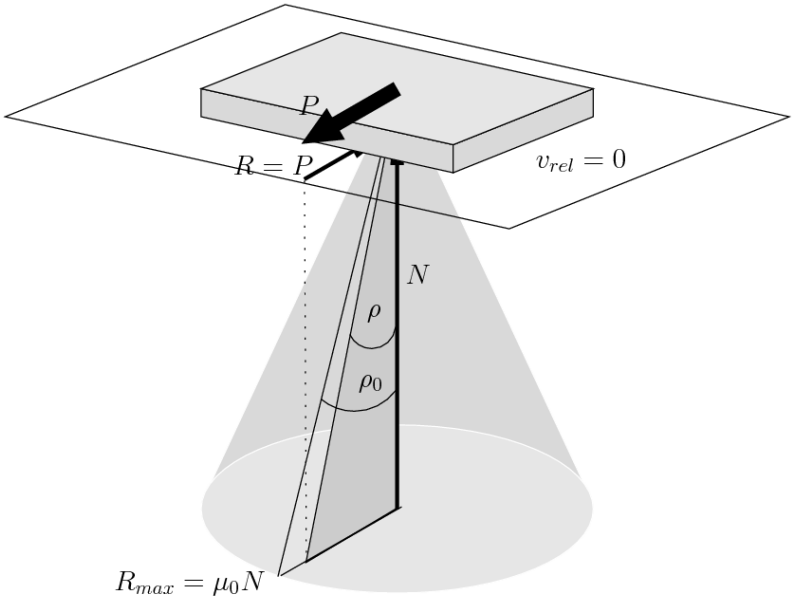
$$|\tan \rho_0| = \tan \rho_0$$

$$\rho \leq \rho_0$$

11.4. Cone of adhesion and angle of adhesion

Geometric interpretation: The resulting attack force must lie within the cone with half the opening angle ρ_0 for the body to adhere.

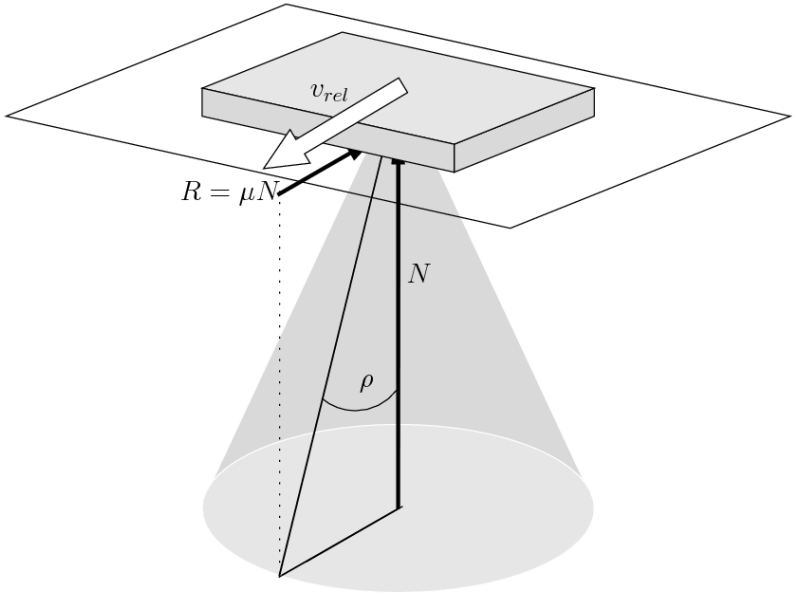
Static Friction



$R_{max} = \mu_0 N$

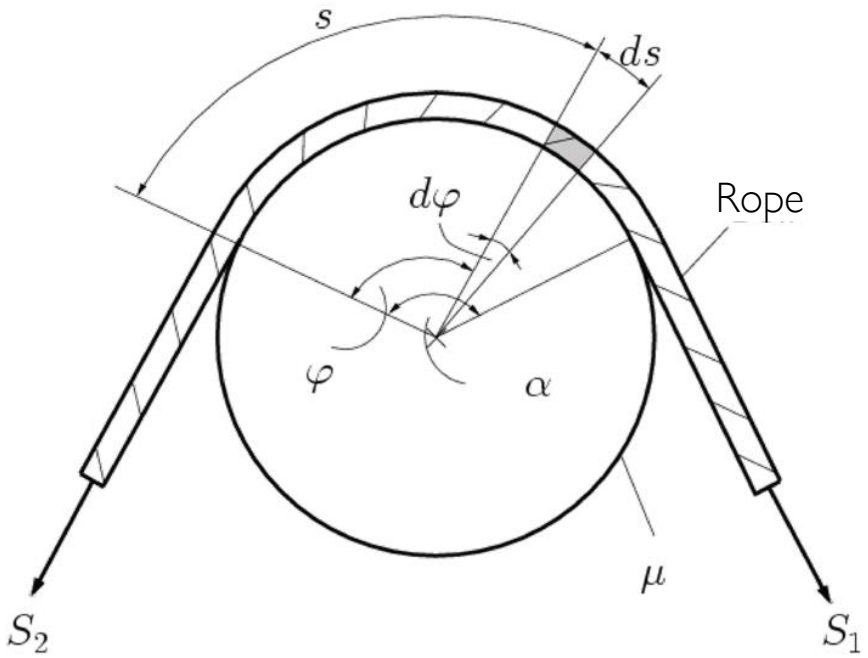
$\rho \leq \rho_0, \tan\rho_0 = \frac{R_{max}}{N} = \mu_0$

Kinetic Friction



$\tan\rho = \frac{R}{N} = \mu$

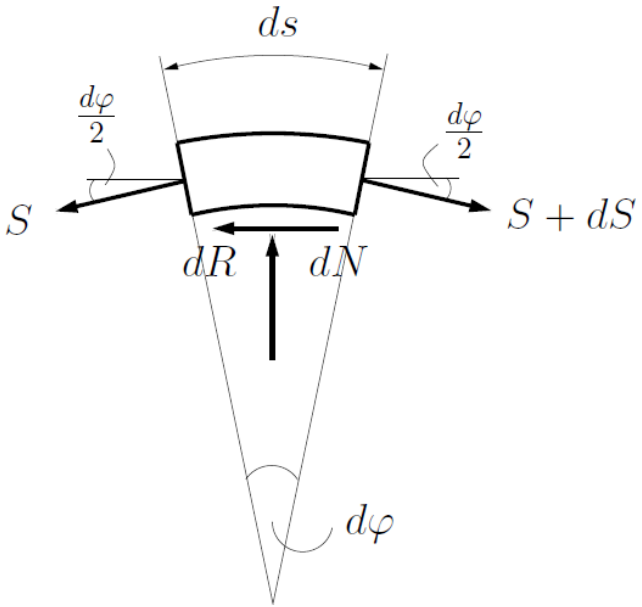
11.5. Belt friction



For which belt forces

- does the belt grip?
- does the belt slide?

11.5. Belt friction - Equilibrium on differential Rope element



- Free cutting and application of free cutting forces (R against potential direction of movement!)
- Equilibrium of forces in normal direction:

Box 11.7

- Equilibrium of forces in tangential direction:

Box 11.8

11.5. Belt friction

- Limit case of maximum adhesive force according to Coulomb's law.

$$dR = \mu_0 dN = \mu_0 \left(S \sin\frac{d\varphi}{2} + (S + dS) \sin\frac{d\varphi}{2} \right)$$

- With small angle approximation and insertion of dR follows:

$$0 = dS - \mu_0 \left(S d\varphi + \underbrace{\frac{1}{2} dS d\varphi}_{\doteq 0 \text{ (2nd Order!)}} \right)$$

- This results in a differential equation with initial condition $S(\varphi = 0) = S_2$

Box 11.9

11.5. Belt friction – Solving diff. equations

- Separation of variables:

$$\frac{dS}{S} = \mu_0 d\varphi$$

- Integration with initial values:

$$\int_{S(0)=S_2}^{S(\varphi)} \frac{d\hat{S}}{\hat{S}} = \mu_0 \int_0^{\varphi} d\hat{\varphi}$$

- Result:

$$\ln\left(\hat{S}\right)\Big|_{\hat{S}=S_2}^{S(\varphi)} = \ln S\left(\varphi\right) - \ln S_2 = \ln \frac{S\left(\varphi\right)}{S_2}$$
$$= \mu_0 \hat{\varphi}\Big|_{\hat{\varphi}=0}^{\varphi} = \mu_0\left(\varphi - 0\right) = \mu_0\left(\varphi\right)$$

- Course of the rope force:

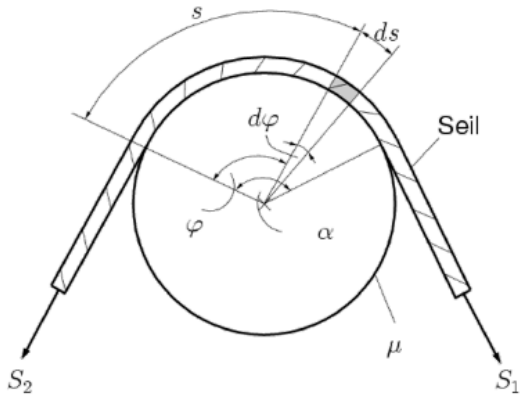
Box 11.10

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11.5. Belt friction - The Euler-Eytelwein formula in static cases

Euler-Eytelwein formula in static case (with $S_1 > S_2$):

- For ϕ=α: static edge case $S_1 = S_2 \cdot e^{\mu_0 \alpha}$
- Static if $S_1 \leq S_2 \cdot e^{\mu_0 \alpha}$
- Analogous derivation for the case $S_1 < S_2$. Together we get:



Box 11.11

11.5. Belt friction

- Sliding if Euler-Eytelwein formulas are not fulfilled in the static case.
- Abrupt transition to sliding friction.
- Sliding friction forces:

$$S_1 > S_2 : S_1 = S_2 e^{\mu\alpha}$$
$$S_1 < S_2 : S_2 = S_1 e^{\mu\alpha}$$

with μ coefficient dependent on material.